CISCaRL

Engineering-Driven Machine Learning Pipeline

Alternatives, experiments, evaluation criteria and recorded decisions for every stage

Version	**2.0 — aligned with the A×E run, checked 7 November 2026**
Scope	ACTG175 arms 0/1, 12 baseline covariates, 730-day composite endpoint
Model	CISCaRL, frozen configuration C200
Baseline	survival causal rule ensemble (SCRE), same splits and horizon
Status	measured findings and proposed experiments are labelled separately

Purpose and standard of evidence

This document is not a description of a model. It is a record of the engineering process that produced it: the alternatives that were considered at each stage, the experiments that were run to separate them, the criteria fixed before looking at the results, the measured findings, and the reasoning that led to the final choice. Where an experiment has not been run, that is stated and the choice is labelled as provisional rather than presented as validated.

The decision loop applied at every stage

- Engineering question — what decision must actually be made here, and at what cost if it is wrong.
- Existing evidence — what the data, the protocol or earlier stages already establish.
- Alternatives — the realistic options, including the one currently implemented.
- Hypothesis — what each alternative is expected to do, and why.
- Experiment — what is varied, what is held constant, and on which data roles.
- Evaluation criteria — fixed before the experiment, and chosen to match the decision.
- Findings — what was actually observed, with uncertainty.
- Decision — what was selected, retained or rejected.
- Reasoning — why the evidence supports that choice rather than the alternatives.
- Limitations — what the experiment could not establish.
- Reproducibility — the seeds, files and configuration that support the conclusion.

Three kinds of decision, kept apart

Type	Definition	Example in this project
Design decision	Justified by the protocol, the estimand, data definitions or a mathematical requirement; an experiment would not change it.	Using arms 0/1; treating censoring as right-censoring; excluding post-baseline columns from X.
Experimentally validated decision	Retained after comparing alternatives on prespecified criteria.	Support floor 200; split 50/25/25; shrinkage for prediction; keeping depth 3 after nested tuning.
Provisional decision	A reasonable initial configuration that has not yet been tested, and is labelled as such.	Stacked nuisances; conformal interval layer; selection-aware inference.

Status labels used throughout

- Completed and executed — the analysis was run and its output is in the repository.
- Completed and loaded — the experiment was run earlier and its saved output is used.
- Proposed, not executed — the protocol exists but no result is claimed.
- Rejected — tested and did not meet the prespecified criterion.

Reading guide

- Stages 1–3 establish the estimand, the audit and the feature set; their evidence is the protocol, the data dictionary and descriptive counts.
- Stages 4–8 are the experimental core: split, nuisance, estimator, learner, selection and commitment. Each is decided by a controlled comparison.
- Stages 9–10 freeze the configuration and then read confirmation seeds and the assessment partition, which is where a claim becomes a result.
- Stages 11–12 compare against SCRE, separate policy value from net benefit, and report which conclusions survive perturbation.
- The registers at the end list every experiment and decision in one place; the reproducibility manifest gives the seeds and files behind each number.

What this document deliberately does not do

- It does not add algorithms to look comprehensive. kNN, SVM, naïve Bayes, clustering, neural networks, PCA, LDA, HMM and reinforcement learning are excluded because they do not produce the interpretable subgroup this project requires.
- It does not present a configuration as validated because it was run. Running is not evidence; comparison on prespecified criteria is evidence.
- It does not hide negative results. The tuning study, the two-leaf list, the flexible learner and the volatility of the selected rule are all reported.

Symbols and abbreviations

Symbol	Meaning	First used
τ(730 | x)	contrast $S_1(730 \mid x) - S_0(730 \mid x)$ for a patient with baseline covariates x	Stage 1
S_a(t | x)	event-free survival probability at t under assignment a	Stage 1
ψ	cross-fitted AIPW pseudo-outcome, one value per discovery patient	Stage 6
AIPW	augmented inverse-probability weighting	Stage 6
IPCW	inverse-probability-of-censoring weights	Stage 5
ESS	effective sample size of the weighted pseudo-outcome	Stage 5
PH	proportional hazards	Stage 5
CART	classification and regression tree (here, regression)	Stage 7
C200	frozen configuration: support floor 200, depth 3, leaf 60, squared error	Stage 7
C200-S	C200 with empirical-Bayes shrinkage used for prediction only	Stage 8
SCRE	survival causal rule ensemble, the single comparator	Stage 11
LCB	one-sided 95% lower confidence bound	Stage 8
MAE	mean absolute error in τ	Stage 11
J / Jaccard	overlap between the selected leaf and the reference region	Stage 11
DGP	data-generating process used in simulation	Stage 4
MCSE	Monte Carlo standard error	Stage 10
ECE	expected calibration error (weighted mean absolute bin gap)	Stage 5
PRIM	patient rule induction method (proposed, not run)	Stage 7

Executive summary

The model is a support-constrained interpretable survival subgroup policy. It estimates each patient's 730-day event-free survival contrast with a cross-fitted AIPW pseudo-outcome, discovers rules with a depth-3 regression tree, selects the highest-scoring leaf that has at least 200 discovery patients, re-estimates that leaf on a disjoint validation partition, and recommends treatment only when the one-sided 95% lower bound is above zero.

Question	Measured answer	Status
Does a discovered rule beat the constant model?	No: 26.4% worse than the constant at n = 2,000; 21.7% better from n ≈ 6,000. The crossing is the operating-range boundary.	Completed
Does the support constraint help?	Capture 0.148 → 0.364 and Jaccard 0.127 → 0.295 in the two-region design; purity rises from 0.77 to 0.82.	Completed
Does calibration close the accuracy gap to SCRE?	Frozen C200 is worse by +0.0062 ($p < 0.001$); after shrinkage +0.0011 ($p = 0.48$) — a tie, not a win.	Completed
Is the method stable?	No: 20 distinct rules from 20 splits, mean Jaccard ≈ 0.34.	Completed, limitation
Does the commitment layer prevent harmful subgroups?	0% null-benefit treatment; IF coverage 90–92.5% (95% confirmation); selection bootstrap rejected.	Completed
Is it better than SCRE?	Only on precision and abstention, and only once treatment has a cost (net benefit leads from λ ≈ 0.05–0.10).	Completed
Does it answer on ACTG175?	No. Largest leaf 173; fallback leaf 61; validation LCB −0.250; INSUFFICIENT.	Completed, power result
Is the inference selection-aware?	No. IF conditions on the rule; the selection bootstrap was tested and rejected (85% / 95% coverage in confirmation, wider than IF).	Tested, rejected

The single most important engineering lesson in the project is the tuning result. The first tuning study reported a large gain for the absolute-error criterion (0.824 versus 0.871). When the nuisances were refitted inside every tuning fold, the gain disappeared (0.8633 versus 0.8638). The first number was an artefact of leakage; the second is the evidence. The correct engineering response was to withdraw the claim, not to keep the better-looking number.

What is experimentally settled — and what is not

The distinction is part of the result, not a disclaimer. A design decision is justified by the protocol or a mathematical requirement; a tested decision is retained after a controlled comparison; a proposed decision has a protocol but no result yet.

Decision	Status	Evidence or reason
Treatment-policy estimand, 730-day horizon	Design + sensitivity	390 off-treatment, 122 subsequent events, follow-up to 1,176 days; 365/730/1,095 change the rule, so the horizon is fixed a priori
12 baseline covariates vs 16 / lasso / RFE	Tested	Gate 38: lasso selects all 12; RFE changes the rule and drops to support 79
730-day horizon	Tested	Gate 44: 730 has the best mean validation LCB and lowest SD across 20 splits
50/25/25 split, A×E strata	Tested	4 ratios × 20 real splits + 20 simulation reps; event-rate spread 0.5 vs 2.8 points
Per-arm Cox + censoring Cox	Design + tested	PH p = 0.235 / 0.368; ECE 0.032. Gate 31: Cox best Brier; stack lower MAE but loses purity and costs 100×+
Cross-fitted AIPW	Design + tested	Audit bias +0.0031; Gate 36: AIPW retained vs TMLE (TMLE MAE 0.08 worse)
Depth-3 CART, leaf 60, squared error	Tested	18 configurations × 5 nested folds; best alternative 0.8633 vs frozen 0.8638
Support floor 200	Tested	Two-region DGP, 40 reps: capture +0.216 [+0.142, +0.291]; 5% floor at n = 2,000
Commit on LCB > 0	Tested	0% null treatment; efficiency 0.84; IF coverage 90–92.5%
Shrinkage for prediction	Tested	80 paired reps; confirmation block and pooled CIs both below zero
Isotonic calibration	Tested, adopted	Gate 33: ΔMAE −0.0053 / −0.0065 on confirmation; policy unchanged
Pruning, PRIM, classification tree	Tested, rejected	Gate 32: none meets the purity guardrail; boosting distillation fails cost

Decision	Status	Evidence or reason
Selection-aware or conformal intervals	Tested	Gate 33 selection bootstrap rejected; Gate 41 Bonferroni and sample-splitting validated; IF retained
Fairness audit, SHAP, internal transport	Tested descriptively	Gate 35: every group under 30 rule patients; no fairness claim
5 cross-fitting folds	Tested	Gate 34: 10 folds improves MAE by 0.0005, below the 0.002 threshold

Stage 1 — Problem definition and estimand

A. Engineering question

What treatment contrast, population, endpoint and horizon are being estimated, and how are post-randomisation events such as treatment discontinuation handled?

B. Why the decision matters

If assignment is confused with treatment received, or discontinuation is treated as an event or as automatic censoring, the analysis answers a different question from the one the trial was designed to answer. Every later stage inherits this choice.

C. Concepts used

Potential outcomes; survival estimands; treatment-effect heterogeneity; the ICH E9(R1) estimand framework and intercurrent events; right-censoring; randomisation; consistency, positivity, exchangeability and non-informative censoring.

D. Evidence available before the stage

- ACTG175 arms 0 and 1, randomised assignment available as the arm indicator.
- Observed time `days`, event indicator `cens` (1 = event observed), post-randomisation `offtrt`.
- Composite progression-free endpoint: CD4 decline ≥ 50, AIDS progression, or death.
- Follow-up extends beyond 730 days: median observed time 989 days, maximum 1,231.

E. Alternatives and trade-offs

Alternative	What it answers	Why it was not chosen
Intention-to-treat with treatment-policy outcomes	Effect of assigning the arm, counting outcomes regardless of discontinuation	Chosen — see decision.
Per-protocol / while-on-treatment	Effect if the assigned drug were actually taken	Requires censoring at discontinuation; 390 patients (37%) discontinue, so this would discard a third of the evidence on a post-randomisation behaviour.
Hypothetical strategy	Effect if everyone had tolerated the drug	Needs an explicit model for what would have happened after discontinuation; not identified without extra assumptions.
Hazard ratio	Relative instantaneous risk	Not the clinical quantity; changes with follow-up length and does not translate into an absolute benefit.
RMST difference	Difference in mean event-free time	Tested earlier (Gate 8) and performed worse than its own constant.

F. Hypothesis

Because follow-up and events continue after discontinuation (122 events among the 390 off-treatment patients, follow-up to 1,176 days), a treatment-policy estimand is the closest match to the observed data and to a decision about what to prescribe at baseline.

G. Experiment and protocol

- Confirm arm coding and the randomisation mechanism; use assignment, not received treatment.
- Verify `cens` coding and the composite endpoint against the data dictionary.
- Quantify follow-up and events in the off-treatment group.
- Prespecify the horizon and the censoring assumption before any model output is examined.

H. Evaluation criteria and decision rule

- Correct arm, time and event coding; no reversed indicator.
- Continued follow-up after discontinuation demonstrated from the data.
- Explicit separation of discontinuation from censoring.
- An explicit statement of the censoring assumption, and an analysis of what would change if it failed.

I. Measurement

Group	n	Events	Censored	Median follow-up	Max follow-up
On treatment	664	162	502	1,062 d	1,231 d
Off treatment	390	122	268	688 d	1,176 d

Measured on the 1,054 analysed patients. Correlation between offtrt and the event indicator: r = 0.075.

J. Decision and reasoning

Adopt the treatment-policy estimand: analyse by original assignment and count outcomes regardless of discontinuation. This is justified by the observed continuation of follow-up after discontinuation. The weaker association with the event indicator does not prove that censoring is independent, and the 375-day shorter follow-up in the off-treatment group is a warning about the observation process rather than proof of informative censoring. The assumption of conditional independent censoring given baseline covariates is therefore stated explicitly and examined by sensitivity analysis.

K. Limitations and stage exit

The estimand is identified only under that censoring assumption. If dropout is driven by unobserved health, baseline IPCW cannot repair it. `offtrt` must not enter the baseline model; an end-of-follow-up indicator cannot predict earlier censoring.

L. Outputs

- Estimand and endpoint specification.
- Off-treatment evidence table.
- Causal-assumption register and censoring sensitivity plan.

Reproducibility. *ACTG175 raw file, arms 0/1 subset; the counts in Table 1 are recomputed in notebook Step 2 and require no seed. The estimand text is the frozen record in `results/CANONICAL_RESULTS.md`.*

Stage 2 — Dataset study and audit

A. Engineering question

What does each of the 27 columns represent, which data-quality issues could change the population or the analysis, and which anomalies should be corrected, excluded, flagged or retained?

B. Why it matters

Role assignment determines which columns may enter the model at all. A cleaning step that silently deletes unusual patients changes the target population; a column measured after randomisation leaks outcome information.

C. Concepts used

Data provenance; data dictionary; EDA; missingness profiling; duplicate detection; variance, sparsity and redundancy; plausibility checks; Kaplan–Meier; censoring; tied event times.

D. Evidence

- Raw file: 2,139 × 27.
- Missingness: only `cd496` (797 of 2,139); the 12 baseline covariates are complete.
- Zero variance: `zprior` is constant in arms 0/1.
- Redundancy: `z30` agrees with `str2` in 94.8% of patients.
- Sparsity: `oprior` is present in 2.4% (25 patients).
- Plausibility: `cd40 = 0` in 3 patients, 199 outside a 200–500 reference range, 13 minors.

E. Alternatives and trade-offs

Alternative	Effect	Decision
Automated cleaning (drop out-of-range, impute everything)	Would delete 199 patients (19%) and invent values for a complete covariate set	Rejected
Flag and retain, with a sensitivity analysis	Preserves the randomised cohort; the anomalies are reported rather than hidden	Adopted
Impute `cd496` and use it	Post-baseline, leaks the endpoint	Rejected
Drop `z30` and `str2` together	Removes two clinically meaningful history variables; only one is redundant	Rejected — keep `str2`

F. Hypothesis

A role-first audit will identify unsafe predictors without changing the population. Unusual values are more likely to be real trial heterogeneity than data errors.

G. Protocol

- Build a 27-row data dictionary: type, role, missing count, unique count, valid range.
- Classify each column: identifier, treatment, outcome, post-baseline, design, baseline.
- Quantify variance, sparsity and redundancy among the 16 baseline candidates.
- Produce distributions, correlation, arm balance and KM curves.
- Quantify tied event times.

H. Criteria and decision rule

Every column has one documented role and one retain/exclude decision; every count is quantified; no exclusion is made because it improves the model.

I. Measurement

Column	Role	Finding	Decision
`pidnum`	identifier	—	exclude
`days`, `cens`, `r`	outcome	—	exclude from X
`arms`, `treat`	treatment	`treat` duplicates `arms`	exclude from X
`cd420`, `cd496`, `cd820`, `offtrt`	post-baseline	`cd496` 37% missing	exclude (leakage)
`strat`	design	derived stratum	exclude
`zprior`	baseline candidate	constant	exclude
`z30`	baseline candidate	94.8% agreement with `str2`	exclude
`oprior`	baseline candidate	2.4% prevalence	exclude
`preanti`	baseline candidate	redundant history	exclude
12 covariates	baseline	complete	retain

Full 27-row audit is in the notebook, Step 2. Counts: 12 retained, 15 excluded (11 by role, 4 by screening).

J. Decision and reasoning

Use role-based exclusion first, then documented redundancy and sparsity checks, to reach the 12 baseline covariates. Retain unusual `cd40` and age values unless a clinical argument supports deletion, and report them.

K. Limitations

The screening uses data from all arms to count prevalence and agreement. That is descriptive; the decision to retain the 12 published covariates is made for comparability, not by an algorithm, and is tested later as a 16-covariate sensitivity.

L. Outputs

- 27-row role audit
- missingness/variance/sparsity report
- EDA figures
- tie summary
- provenance manifest

Reproducibility. *notebook Steps 2–3; the audit is a descriptive pass over the raw file, so no seed is involved. The 27-row table is reproduced verbatim in the notebook output.*

Stage 3 — Cohort and feature selection

A. Engineering question

Which patients and which baseline characteristics form the analysis population, and can each choice be defended independently of the result it produces?

B. Why it matters

The cohort defines the population to which the result applies; the feature set defines which rules can be expressed at all.

C. Concepts

Cohort construction; inclusion/exclusion; baseline versus post-baseline; leakage; redundancy screening; clinical interpretability; feature selection versus feature extraction.

D. Evidence

- Arms 0/1: 1,054 patients (532 control, 522 treated).
- 284 events (26.9%); 103 censored before 730 days (9.8%); 750 at risk at 730 (71.2%).
- The 12 covariates are the published Tsiatis set, also used by SCRE.

E. Alternatives

Alternative	Consequence	Decision
All 27 columns	Includes outcome, treatment and post-baseline columns — leakage	Rejected
16 candidates: the 12 plus the four screened columns	Stress test that re-admits `oprior` (2.4%), `z30` (94.8% duplicate), `preanti` and the design column `strat`	Tested as sensitivity
12 published covariates	Comparable with SCRE; all complete	Adopted
Data-driven selection (lasso/RFE)	Drops variables after seeing effects; breaks comparability	Rejected for the primary model
PCA / LDA components	Rules become uninterpretable	Rejected

F. Hypothesis

The four screened columns add no usable information to the selected rule: one is constant, one duplicates `str2`, one is too sparse, and one is a design stratum. Re-admitting them should therefore not change the rule; if it does, the covariate screening is driving the result.

G. Protocol

- Restrict to arms 0/1 and document the counts.
- Apply the 27-row role decision.
- Fit the primary model on 12 features.
- Refit with 16 features as a sensitivity and compare the selected rule.

H. Criteria

The primary feature set must preserve interpretability and comparability; any alternative is adopted only if it changes the rule materially and improves a prespecified metric.

I. Findings

Analysis	Rule selected	Support	Conclusion

Analysis	Rule selected	Support	Conclusion
12 covariates (primary)	`cd40 ≤ 407.5 & wtkg > 71.0 & homo ≤ 0.5`	61	Reference
16 columns (sensitivity: the four screened ones re-added)	identical rule	61	The feature restriction does not drive the rule

J. Decision and reasoning

Retain the 12 covariates. The sensitivity shows the selected rule is unchanged when the four screened variables are restored, so comparability is obtained without cost. PCA and learned feature selection remain excluded because they would make the rules uninterpretable or post hoc.

K. Limitations

The equality of the 12- and 16-column rules is one split of 1,054 patients, not a distribution. It is evidence of stability to the feature set, not proof. The sensitivity also re-admits `strat`, a design column, so it is a stress test of the screen, not a candidate baseline model.

L. Outputs

- cohort flow
- 12-feature matrix
- 27-row audit
- 16-covariate sensitivity

Reproducibility. *notebook Steps 3–6 and the sensitivity cell (discovery seed 2024); the 16-column refit uses the same split and the same cross-fitting seed as the primary model.*

Stage 4 — Splitting and preprocessing

A. Engineering question

How much data should discovery, validation and assessment receive, how should the split be stratified, and which transformations must be fitted on discovery only?

B. Why it matters

Discovery must be large enough to find a rule; validation must be large enough to decide whether to trust it; assessment must be untouched. The ratio trades these three against each other, and every fitted transform is a leakage risk.

C. Concepts

Train/validation/test separation; stratification; event-rate balance; leakage; standardisation; centring and the Cox baseline; reproducibility.

D. Evidence

- 1,054 patients, 284 events.
- Stratifying by treatment alone gives event rates 26.0 / 27.0 / 28.8 across partitions (spread 2.8 points).
- Stratifying by treatment × event gives 27.1 / 26.6 / 26.9 (spread 0.5 points).

E. Alternatives

Ratio	Discovery	Validation	Real: median val SE	Real: commit	Sim: purity	Sim: capture
40/30/30	421	316	0.147	30%	0.868	0.378
50/25/25	527	263	0.157	25%	0.941	0.479
60/20/20	632	211	0.126	5%	0.94	0.432
70/15/15	737	158	0.142	30%	0.965	0.463

20 real-data splits and 20 simulation replications per ratio. Every ratio produced 20 distinct rules from 20 splits.

F. Hypothesis

A larger discovery set improves the rule, but the ratio acts mainly through which leaf the support floor forces, not through raw precision.

G. Protocol

- Fix the split before any fitted transform.
- Stratify by treatment × event.
- Fit the scaler on discovery and apply it unchanged.
- Test alternative ratios on separate development runs, never on the final assessment.

H. Criteria

Retain the ratio with the best combination of capture, purity and commitment. Do not select on assessment performance. Do not claim mathematical optimality.

I. Findings

60/20/20 has the smallest median validation SE (0.126) but the lowest commitment (5%). The larger discovery set clears the 200-patient floor in 12 of 20 splits (median support 207.5), but those larger leaves carry smaller effects (mean validation estimate +0.082 versus +0.116 for sub-floor leaves), so the median lower bound is the most negative (−0.110). 40/30/30 commits in 30% of splits but falls back in all 20, and loses 0.10 capture and 0.07 purity

in simulation. 50/25/25 gives the highest simulated capture (0.479) at purity 0.941, but in the real splits it falls back in 80% and commits in only 25%. 70/15/15 has the highest simulated purity (0.965) and the same 30% commitment, but lower capture (0.463).

J. Decision and reasoning

Retain 50/25/25 with treatment × event stratification. The evidence is that the ratio should be chosen on the discovery side, because it determines the rule; a larger discovery set buys a bigger, weaker leaf rather than a better decision. The choice is a documented compromise, not an optimum, and the rule instability (20/20 distinct rules) is reported as a limitation of the data size.

K. Limitations

20 replications per ratio support a directional finding, not significance. Only one DGP was simulated for the ratio comparison.

L. Outputs

- split table with event rates
- split-ratio study
- discovery-fitted scaler
- leakage checklist

Reproducibility. *split seeds 2024 (discovery vs holdout) and 2025 (validation vs assessment); ratio study seeds 4000–4019 for the 20 real splits per ratio, with the simulation block recorded separately in `results/split_study_real.csv` and `results/split_study_sim.csv`.*

Stage 5 — Survival and nuisance models

A. Engineering question

Which specification produces survival and censoring estimates accurate enough for AIPW, and how do we know?

B. Why it matters

AIPW is doubly robust, but only one nuisance needs to be wrong to lose that protection. A model that predicts survival slightly better does not automatically produce a better causal contrast.

C. Concepts

Cox PH; baseline survival; Efron/Breslow ties; conditional censoring; IPCW; weight truncation; effective sample size; PH diagnostics; survival calibration.

D. Evidence

- 12 covariates, 527 discovery patients, 284 events overall.
- Censoring before 730: 10.4% in discovery.

E. Alternatives

Model	Strength	Risk	Status
Cox PH (adopted)	Parsimonious; interpretable; stable with 12 covariates	PH assumption; linear predictor	Completed
Random Survival Forest	Captures nonlinearity and interactions	Gate 1c: AIPW-RSF indistinguishable from AIPW-Cox when the within-arm Cox is correct; as a plug-in target it was worse than the constant (Gate 9)	Tested, not adopted
Survival boosting	Strong predictive fit	Tuning burden; opacity	Tested (Gate 31)
Stacking	Combines strengths	Needs nested fitting; can hide instability	Tested (Gate 31)

F. Hypothesis

Global PH is likely adequate; if one covariate violates, that is a reason to test a flexible nuisance rather than to discard Cox.

G. Protocol

- Fit per-arm Cox and a conditional censoring Cox on discovery with 5-fold cross-fitting.
- Run Schoenfeld tests per arm; compare Efron and Breslow.
- Calibrate predicted S(730) against KM in risk bins on assessment.
- Sweep the weight cap and report how many weights bind.

H. Criteria

Retain Cox if global PH is not rejected, calibration is acceptable, weights are stable, and the downstream AIPW audit passes. Adopt a flexible model only if it improves downstream estimation without destabilising weights.

I. Findings

Check	Result	Reading
PH global (Fisher, rank)	p = 0.235 (arm 0), 0.368 (arm 1)	No global violation
PH per covariate	`cd40` 0.021 / 0.023; `str2` 0.022; `wtkg` 0.048	`cd40` in both arms; also `str2` (arm 0), `wtkg` (arm 1)

Check	Result	Reading
Efron vs Breslow	max coefficient difference 0.0014	Tie handling immaterial
Calibration at 730	lowest-risk bin gap −0.006 / −0.013; ECE 0.032	Good absolute calibration; bin gaps up to 0.084
Folds / IPCW ESS	106/106/105/105/105; ESS 372 of 527 (71%)	Usable information retained
Weight cap	binds only at 1.2 (9.9%); ≥1.5 never binds	Cap choice is not driving the result

J. Decision and reasoning

Retain the per-arm Cox and conditional censoring Cox. The global PH assumption holds, the probabilities are calibrated, and the weights are benign. RSF was tested as an AIPW nuisance in Gate 1c and did not improve on Cox when the within-arm Cox was correct; as a plug-in target it was worse than the constant (Gate 9). Survival boosting and stacking were tested in Gate 31 but did not dominate, so they remain non-adopted. The covariate-level violations (`cd40` in both arms, `str2` in arm 0, `wtkg` in arm 1) are recorded as the specific reason a full misspecification comparison remains legitimate future work.

K. Limitations

The calibration check is on 264 assessment patients, so bin gaps carry substantial uncertainty. The PH test has limited power at 527 patients.

L. Outputs

- diagnostics tables
- PH plots
- calibration table
- cap sweep

Reproducibility. *cross-fitting seed 2024 on the discovery partition; outputs are `results/diagnostics.md`, `diagnostics_ph.csv`, `diagnostics_calibration.csv`, `diagnostics_caps.csv` and `diagnostics_ties.csv`. The cap sweep changes only the cap, holding folds, nuisances and ψ fixed.*

Stage 6 — Cross-fitting and AIPW

A. Engineering question

Which estimator turns the nuisances into a patient-level treatment-effect target, and does it survive the leakage and stability checks?

B. Why it matters

The tree learns from the pseudo-outcome. If the pseudo-outcome is biased or overfit, every later stage inherits the error.

C. Concepts

AIPW; IPW; influence functions; Neyman orthogonality; cross-fitting; nested refitting; weight truncation; ESS.

The estimator is the cross-fitted augmented inverse-probability-weighted contrast: $\psi_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + A_i(Y_i - \hat{\mu}_1(X_i))/\hat{e} - (1 - A_i)(Y_i - \hat{\mu}_0(X_i))/(1 - \hat{e})$, where $\hat{\mu}_a$ is the arm-specific 730-day survival estimate fitted without patient i, Y_i is the observed event-free indicator at 730, and $\hat{e} = 0.5$ by randomisation.

D. Evidence

- e = 0.5 by randomisation (532 vs 522).
- 5 folds; cap min(Q0.99, 20).

E. Alternatives

Estimator	Protection	Weakness	Status
AIPW (adopted)	Consistent if either nuisance is right	Needs both to be approximately right for efficiency	Completed
Pooled Cox / Cox T-learner	Low variance; simple	Gate 1c: bias 16–45% of τ under crossing hazards, sign flips with censoring	Tested, rejected
Plug-in only	Simple	Gate 9/17: worse MAE than the constant; region gain cancelled by null-region variance	Tested, rejected
IPW only	Outcome-model free	High variance; unstable weights (Gate 1)	Tested, rejected
TMLE	Doubly robust, targeted	Gate 36: MAE 0.079–0.089 worse, Brier worse, bias larger, cost 4.5 s	Tested, rejected

F. Hypothesis

Cross-fitted AIPW will be nearly unbiased, and its pseudo-outcome will be centred on the marginal contrast with substantial patient-level variance.

G. Protocol

- Cross-fit nuisances over 5 folds; build psi out-of-fold.
- Inspect mean, sd, extremes; audit bias against known truth in simulation.
- For tuning, refit nuisances inside every inner fold.

H. Criteria

$|\text{bias}| < 0.01$ in the audit; ψ mean close to the marginal estimate; weights stable; no leakage path between inner-validation rows and inner-training pseudo-outcomes.

I. Findings

Check	Result
ψ mean / sd	+0.180 / 0.997
AIPW bias (audit)	+0.0031 (sd 0.0130)
Arm means vs observed	within 0.002
Nested refit test	naive tuning 0.824 vs nested 0.8638 — leakage confirmed

J. Decision and reasoning

Adopt cross-fitted AIPW with e = 0.5. The audit shows negligible bias, and the nested comparison demonstrates why the inner loop must refit the nuisances: without it, tuning rewards configurations that exploit leakage. Pooled Cox and the Cox T-learner had lower RMSE in Gate 1c, but their bias under crossing hazards was 16–45% of τ and its sign changed with censoring. For a patient-level contrast, that bias is disqualifying; RMSE alone is not the decision criterion.

K. Limitations

Double robustness is a large-sample property and does not protect against unmeasured confounding, informative censoring or a mis-specified endpoint. The number of folds was fixed at five as a design choice; 3-fold and 10-fold variants were not compared, so the report does not claim that five is optimal.

L. Outputs

- ψ diagnostics
- bias audit
- nested-tuning comparison
- weights/ESS table

Reproducibility. *5-fold cross-fitting with seed 2024; the bias audit is the 10-replication n = 2,000 run in `results/AUDIT_report.md`; nested tuning is refitted on the real discovery split (seeds 2024/2025) and recorded in `results/nested_tuning.csv`, so it uses no simulation seed block.*

Stage 7 — Subgroup discovery

A. Engineering question

Which rule-producing learner, and which complexity settings, give the best subgroup at an acceptable level of interpretability?

B. Why it matters

A flexible model can fit ψ better and still produce worse subgroups. The decision must be made on subgroup and policy metrics, not on training fit.

C. Concepts

Regression CART; recursive partitioning; depth; minimum leaf size; squared vs absolute error; pruning; rule extraction; distillation; PRIM.

D. Evidence

- 6 leaves at depth 3; supports 61–173; ψ from −0.184 to +0.556.
- Alternatives already tested in earlier gates: two-leaf list (purity 0.862 → 0.696), flexible learner + distillation (capture 0.548 → 0.454, purity 0.829 → 0.897).

E. Alternatives

Learner / setting	Expected effect	Measured	Decision
Depth 3 / leaf 60 / squared (frozen)	Interpretable, stable	0.8638 nested CV	Retained
Depth 2	Sparser rules, less capture	0.8672	Rejected
Depth 4	Finer rules, lower purity	0.8633	Tie; rejected for interpretability
Depth 3 / leaf 40	Finer leaves, more variance	0.8793	Rejected
Depth 3 / leaf 100	Coarser leaves, less capture	0.8721	Rejected
Absolute-error criterion	Appeared better (0.824) on naive CV	0.8711 nested	Claim withdrawn
Pruned CART (`ccp_alpha`)	Simpler tree	capture +0.111 single (ns), +0.261 two; purity −0.149 / −0.173	Tested, rejected
PRIM	Box-shaped subgroup	capture −0.386 / −0.174; purity −0.561 / −0.338	Tested, rejected
Flexible DR-learner + distillation	Purer but narrower	capture 0.548 → 0.454; purity 0.829 → 0.897	Rejected
RF / boosting distillation	Nonlinear survival fit	capture +0.012 / −0.092; purity −0.006 / −0.050; 25–105 s	Tested, rejected (cost)
Gini/entropy tree on sign(ψ)	Classification variant	capture +0.045 / +0.024; purity −0.219 / −0.110	Tested, rejected

F. Hypothesis

A shallow tree is adequate; depth and leaf size matter less than the leakage-free construction of the target.

G. Protocol

- Tune depth, leaf size and criterion by inner CV, refitting nuisances inside every fold.
- Compare with the frozen configuration on the same folds.
- Adopt only if the prespecified CV criterion improves and the rule stays interpretable.

H. Criteria

Lower nested CV MAE, no loss of purity, at most three conditions per rule, and a stable choice across folds.

I. Findings

Configuration	Nested CV MAE	Fold sd
depth 4 / leaf 60 / squared	0.8633	0.0227
depth 3 / leaf 60 / squared (frozen)	0.8638	0.0208
depth 3 / leaf 60 / absolute	0.8711	0.0204

5 outer folds, nuisances refitted inside each. The spread is smaller than one fold standard deviation.

J. Decision and reasoning

Keep depth 3, minimum leaf 60 and the squared-error criterion. The best alternative is 0.0005 better, which is noise. The absolute-error advantage reported earlier is withdrawn: it came from tuning on a ψ computed once on the whole discovery set. Pruning and PRIM were tested in Gate 32 and rejected on purity, capture or cost.

K. Limitations

Only CART, the two-leaf rule list and the flexible DR-learner were compared on the same protocol; PRIM, pruning and a classification tree were tested in Gate 32 and rejected. The nested CV uses 5 folds on 527 patients.

L. Outputs

- nested-tuning table
- leaf table
- rule text
- decision record

Reproducibility. *18 configurations × 5 outer folds on the real discovery partition (seeds 2024/2025); nuisances refitted inside every fold. Evidence: `results/nested_tuning.csv`; the two rejected learners are in `results/gate25_summary.csv` and `results/gate26_summary.csv`.*

Stage 8 — Rule selection, commitment and calibration

A. Engineering question

Which candidate rule is selected, how is its effect estimated independently, and under what condition is treatment recommended?

B. Why it matters

The highest discovery score is not evidence of benefit. Without an independent estimate and a decision rule, the method reports a winner's-curse effect.

C. Concepts

Support constraint; winner's curse; validation-based estimation; influence-function SE; one-sided LCB; commitment and abstention; shrinkage; selection-aware inference.

D. Evidence

- No leaf reaches 200 discovery patients (largest 173).
- Selected leaf: `cd40 ≤ 407.5 & wtkg > 71.0 & homo ≤ 0.5`, support 61.

E. Alternatives

Design choice	Alternatives	Evidence	Decision
Support floor	max mean (no floor) / 100 / 200 / 400 / proportional	two-region DGP, n = 8,000, 40 reps: no floor capture 0.148, purity 0.77; 200: 0.364 / 0.82; 400: 0.441 / 0.78. C200 vs no floor: capture +0.216 [+0.142, +0.291]. At n = 2,000 the 5% floor beats absolute 200 (purity 0.73 vs 0.47, capture 0.127 vs 0.096)	200 at n ≥ 4000; proportional below
Decision rule	always treat / LCB > 0	0% null treatment; efficiency 0.82–0.88	LCB > 0
Calibration	none / shrinkage / isotonic	Gate 33: isotonic ΔMAE −0.0053 (single), −0.0065 (two) on confirmation; shrinkage not significant. Policy unchanged	isotonic, prediction only (adopted)
Inference	IF / validation bootstrap / selection bootstrap / conformal	Gate 33 confirmation: IF 95%; selection bootstrap 85% (single) / 95% (two); conformal 100% but 2.33× width	IF interval retained; selection bootstrap rejected

F. Hypothesis

A support-constrained leaf will be large enough to estimate, and the validation lower bound will reject small leaves.

G. Protocol

- Select on discovery with the support floor; flag fallback.
- Re-estimate the frozen rule on validation; compute SE and LCB.
- Commit only if LCB > 0; otherwise INSUFFICIENT.
- Apply shrinkage to predictions only, never to the rule or the decision.

H. Criteria

Purity must stay within 0.05 of the unconstrained rule (the pre-registered Gate 23 criterion); null-treatment rate must stay at 0%; commitment must not fall; the calibration must not change the decision.

I. Findings

Quantity	Value
Discovery support	61 (11.6%)
Validation effect	+0.116 (SE 0.222), n = 31
One-sided LCB	−0.250
Decision	INSUFFICIENT
Assessment, rule	+0.401 [+0.033, +0.770], n = 38
IF coverage in simulation	90% (Gate 33) / 92.5% (policy audit), width 0.20–0.27

J. Decision and reasoning

Use support 200 with the documented fallback, and commit only on a positive lower bound. On ACTG175 the floor cannot bind and the validation interval excludes commitment, so the correct output is abstention. Isotonic calibration is the confirmed prediction calibration (C200-I); C200-S (shrinkage) is the version used in the SCRE head-to-head, not the final prediction calibration.

K. Limitations

The IF interval conditions on the selected rule; its measured coverage is 90–92.5% in the policy audit and 95% in the Gate 33 confirmation. The selection bootstrap was rejected: it covered 85% in the single-region confirmation and 95% in the two-region, both wider than IF. Split-conformal covered 100% but was 2.33× wider than IF and was rejected on width.

L. Outputs

- selection table
- commitment arithmetic
- assessment table
- coverage audit

Reproducibility. *split seeds 2024/2025; support-floor sweep in `results/gate7_support_sweep.csv` and Gate 23; shrinkage and coverage in `results/gate30_confirm_summary.csv`; the assessment read is notebook Step 23 and `results/CANONICAL_RESULTS.md`.*

Stage 9 — Freeze the methodology

A. Engineering question

What exactly is frozen before the confirmatory read, and where is it recorded?

B. Why it matters

If any choice can change after the assessment is seen, the assessment is no longer independent.

C. Concepts

Configuration management; pre-registration; decision log; reproducibility; version control.

D. Evidence

- 16 recorded decisions; two were result-informed and are labelled as such (split criterion and shrinkage).

E. Alternatives

Approach	Risk	Decision
Freeze only the model file	Preprocessing and thresholds drift	Rejected
Freeze the full configuration and log every decision with its evidence	More documentation	Adopted
No freeze; report the best run	Assessment becomes development data	Rejected

F. Protocol

Record cohort, features, preprocessing, nuisances, cross-fitting, estimator, learner, hyperparameters, support, calibration, decision rule, seed blocks, code version and the canonical result record. The frozen object is the whole table below, not only the model file.

Layer	Frozen value
Estimand	$\tau(730 \mid x) = S_1 - S_0$; composite progression-free endpoint; treatment-policy strategy
Population	ACTG175 arms 0/1; n = 1,054; 532 control / 522 treated
Features	12 baseline covariates (cd40, cd80, age, wtkg, karnof, hemo, homo, race, gender, drugs, str2, symptom)
Partitions	527 / 263 / 264; stratified by treatment × event; seeds 2024, 2025
Preprocessing	mean and sd fitted on discovery only; Cox baseline centred on discovery
Nuisances	per-arm Cox + conditional censoring Cox; 5 folds; seed 2024; cap min(Q0.99, 20)
Target	AIPW ψ; e = 0.5 by randomisation
Learner	depth-3 regression CART; min_samples_leaf 60; squared_error
Selection	max discovery mean ψ among leaves with ≥ 200 discovery patients; fallback flagged
Commitment	validation one-sided 95% LCB > 0; otherwise INSUFFICIENT
Calibration	empirical-Bayes shrinkage of leaf effects for prediction only (C200-S)
Evidence record	`results/CANONICAL_RESULTS.md` and `results/CLAIMS_v1.17.md`
Code	`notebooks/CISCaRL_31_steps.ipynb` + `ciscarl2/`

G. Criteria

Every number in the paper must be traceable to the canonical record, and the frozen configuration must reproduce it from the raw file.

H. Findings

The paper previously reported the older treatment-only split while the notebook used the A×E split. The two disagreed on the selected rule and on the effect. The paper, tables, manuscript and claims have now been regenerated from the A×E run; `results/CANONICAL_RESULTS.md` is the single source of truth.

I. Decision and reasoning

Freeze the A×E configuration and treat any later change as a new experiment, not an edit. The paper–notebook mismatch is resolved by regeneration, not by choosing whichever number looked better.

J. Limitations

Assessment was read for descriptive reporting before the freeze was formalised. No modelling choice used it; that exposure is disclosed, and a fresh split would be a new experiment rather than a repair.

K. Outputs

- configuration record
- decision log
- canonical results
- claims file

Reproducibility. *the frozen configuration is printed in notebook Step 19; the 16 decisions and their evidence are the `DECISIONS` table in the same step; the canonical record is `results/CANONICAL_RESULTS.md` and the defensible claims are `results/CLAIMS_v1.17.md`.*

Stage 10 — Confirmation and final assessment

A. Engineering question

Does the frozen procedure reproduce on seeds it has never seen, and what does it achieve on the untouched assessment partition?

B. Why it matters

A gain on tuning seeds is a proposal. Only fresh seeds convert it into a result.

C. Concepts

Confirmatory analysis; seed blocks; Monte Carlo error; held-out assessment; exposure audit.

D. Evidence

- Gate 28 head-to-head 300100–300139; Gate 30 tuning 310100–310139; Gate 30 confirmation 320100–320139; policy evaluation 700100. The 1,000-replication sweeps are descriptive and carry no confirmatory seed block.

E. Alternatives

Approach	Weakness	Decision
Report the tuning seeds	Optimistic	Rejected
Fresh-seed confirmation then assessment	More compute	Adopted
Assessment used during development	No independent read	Rejected

F. Protocol

- Re-run the frozen configuration on fresh seeds.
- Report Monte Carlo standard errors for every mean.
- Compute every assessment quantity once, at Step 23, after the freeze.

G. Criteria

Direction must hold; a change is promoted only if its paired CI excludes zero, purity falls by no more than 0.02, the null-treatment rate stays 0%, commitment does not fall, and capture does not fall by more than 0.02. The criteria were fixed in `results/gate30_PREREGISTRATION.md` before the confirmation block was run.

Criterion fixed before confirmation	Result	Verdict
MAE improves; paired 95% CI below zero	single −0.0047 [−0.0089, −0.0006]; two −0.0027 [−0.0057, +0.0003]; pooled −0.0051 / −0.0026	Pass single; directional two
Purity falls by no more than 0.02	0.8719 / 0.8777 unchanged	Pass
Null-treatment rate stays 0%	0% in both blocks	Pass
Commitment does not fall	72.5% / 62.5% unchanged	Pass
Capture does not fall by more than 0.02	0.4510 / 0.2562 unchanged	Pass
Absolute-error criterion	0.8711 nested vs 0.8638 frozen	Fail — withdrawn

H. Findings

Block	Result
Shrinkage, single region	ΔMAE −0.0047 (p = 0.022); pooled −0.0051
Shrinkage, two regions	ΔMAE −0.0027 (p = 0.078); pooled −0.0026

Block	Result
IF interval coverage	92.5% (policy audit); 90% (Gate 33)
MCSE (purity, 1,000 reps)	≈ 0.005–0.006 per method

I. Decision and reasoning

Retain shrinkage as a confirmed prediction calibration, with the two-region effect reported as directionally consistent but not individually significant. Do not promote the absolute-error criterion, which failed its confirmation.

J. Limitations

Forty seeds bound the Monte Carlo error only loosely; differences below about 0.005 in MAE should not be interpreted.

K. Outputs

- confirmation table
- MCSE report
- assessment table
- exposure log

Reproducibility. *confirmation seeds 320100–320139, disjoint from the tuning block 310100–310139; policy evaluation uses seed block 700100; the 1,000-replication sweeps are descriptive. Outputs: `results/gate30_confirm_summary.csv`, `sweep1000_two_summary.csv`, `sweep1000_stress_summary.csv`.*

Stage 11 — Evaluation, policy usefulness and SCRE comparison

A. Engineering question

What does the method do well or poorly against known truth and against the comparator, and does the resulting policy provide value beyond accuracy?

B. Why it matters

A method can be less accurate per patient and still be a better policy; collapsing the two into one score hides the trade.

C. Concepts

MAE; RMSE; bias; purity; capture; Jaccard; coverage; per-treated benefit; commitment; interval coverage; policy value; net benefit; paired comparison; Holm correction.

D. Evidence

Semi-synthetic DGPs with analytic truth; 80 paired replications for the head-to-head and 1,000 for the sweeps; ACTG175 assessment.

E. Primary and secondary criteria

Dimension	Primary	Secondary
Prediction	MAE	RMSE, bias
Subgroup	purity at matched coverage	capture, Jaccard
Inference	interval coverage	width
Decision	null-treatment rate	commitment, efficiency
Policy	net benefit at a stated λ	policy value V(d)

F. Findings — head-to-head (80 paired reps)

Design	Method	MAE	vs SCRE	purity	capture	per-treated
single	C200	0.0576	+0.0062	0.890	0.465	+0.174
single	C200-S	0.0525	+0.0011	0.890	0.465	+0.174
single	SCRE	0.0514	—	0.437	0.954	+0.085
two	C200	0.0707	+0.0031	0.858	0.277	+0.165
two	C200-S	0.0680	+0.0004	0.858	0.277	+0.165
two	SCRE	0.0676	—	0.599	0.823	+0.101

G. Findings — policy value and net benefit (40 paired reps)

Design	none	C200	SCRE	treat all
single	0.7659	0.7870	0.7926	0.8145
two	0.7659	0.7855	0.8044	0.8345

Net benefit U = V − λ·treat_rate: at λ = 0.05 the single-region values are C200 0.7810, SCRE 0.7651 and treat-all 0.7645; at λ = 0.10 the two-region values are C200 0.7735, SCRE 0.7484 and treat-all 0.7345. C200 leads once a treated patient costs more than about 0.05–0.10 survival-probability units.

H. Findings — ACTG175

Quantity	Value	Status
Overall τ(730)	+0.124 [−0.002, +0.250]; one-sided LCB +0.018	Measured
Selected rule	`cd40 ≤ 407.5 & wtkg > 71.0 & homo ≤ 0.5`, support 61	Measured
Validation	+0.116 (SE 0.222), LCB −0.250	Measured
Decision	INSUFFICIENT	Measured
Assessment, rule	+0.401 [+0.033, +0.770], n = 38	Descriptive
KM check	+0.341 [+0.177, +0.501]; marginal +0.133	Descriptive

I. Decision and reasoning

Report the two methods as different operating points, not as a ranking. SCRE is more accurate per patient and treats more people; C200 is purer, roughly twice as beneficial per treated patient, and can abstain. On ACTG175 the support constraint cannot be met, so the honest output is abstention. No accuracy claim is made against SCRE.

J. Limitations

Metrics requiring truth exist only in simulation. The comparator uses one α path that retains about one rule. Policy value depends on the assumed cost λ.

K. Outputs

- head-to-head table
- sweep tables
- policy and net-benefit tables
- ACTG175 table

Reproducibility. *80 paired replications = seeds 300100–300139 × two DGPs for the head-to-head; 1,000 descriptive replications per design for the sweeps; 40 paired replications at seed 700100 for policy value. Evidence: `results/gate30_*.csv`, `results/policy_eval_summary.csv`, `results/net_benefit.csv`, and the ACTG175 rows of `CANONICAL_RESULTS.md`.*

Stage 12 — Robustness, fairness, explainability and delivery

A. Engineering question

Which conclusions survive reasonable perturbations, and which are specific to one design choice?

B. Why it matters

A result that changes with the horizon, the feature set or 5% missingness is a property of the analysis, not of the treatment.

C. Concepts

Sensitivity analysis; ablation; censoring tilt; fairness definitions; group calibration; SHAP; reproducibility.

D. Evidence and findings

Perturbation	Result	Conclusion
Horizon 365 / 730 / 1,095	Rule and effect both change; discovery $\hat{\tau}$ +0.062 / +0.180 / +0.162; 7 / 6 / 5 leaves	Horizon-specific; 730 fixed on clinical and support grounds
16 columns (four screened re-added)	Identical rule and support	The feature screen is not driving it
5% missingness in `cd40`	Rule changes to `cd80 > 805.0 & race > 0.5`	The rule is not robust to missingness
Weight cap	Binds only at 1.2	Cap choice is not driving it
δ-tilt	+0.401 to +0.456, intervals above zero	Direction stable, intervals wide
Split ratio	20/20 distinct rules at every ratio	Rule instability is a data-size limit

The acceptance standard for robustness was fixed before the perturbations were run: a choice is robust if the selected rule is unchanged, or if the effect keeps its sign and the interval stays on the same side of zero; otherwise the dependence is reported.

Perturbation	Robust if	Observed	Verdict
Horizon	Direction does not reverse	$\hat{\tau}$ +0.062 / +0.180 / +0.162, all positive	Direction robust; rule changes
Feature set	Same rule and support	Identical on 12 and 16 columns	Robust
Missingness (5% in cd40)	Same rule	Rule becomes `cd80 > 805.0 & race > 0.5`	Not robust
Weight cap	Rule unchanged above 1.5	Unchanged; cap binds only at 1.2	Robust
Censoring tilt	Interval stays above zero	+0.401 to +0.456, all intervals above zero	Robust in direction
Split ratio	Jaccard above 0.5	20/20 distinct rules; Jaccard ≈ 0.34	Not robust — data-size limit

E. Alternatives

Analysis	Value	Status
Bootstrap stability by own-procedure refits	Jaccard ≈ 0.34	Completed
Group-wise calibration and commitment by race/gender	All groups < 30 rule patients; intervals overlap	Completed, descriptive
SHAP on a surrogate for ψ	wtkg 0.048, cd40 0.046, cd80 0.042; supplementary	Completed
External validation (PBC)	Out of scope: ACTG175 only	Internal white→non-white transport completed; support 74 < 200

F. Decision and reasoning

Report every perturbation and its consequence. Do not tune the model until a favourable stability number appears: the honest finding is that the rule is sensitive to the sample and to missingness, which is a property of 527 discovery patients, not a fixable defect.

G. Limitations

Fairness, SHAP and internal transport were executed in Gate 35; they are descriptive and underpowered, not confirmatory. External validation on PBC was not attempted because the project is scoped to ACTG175 only; the white→non-white transport check is the internal substitute.

H. Outputs

- ablation table
- stability report
- sensitivity figures
- limitations section

Reproducibility. *notebook Steps 24–25; horizon, missingness, 16-column and δ-tilt cells use the same discovery seed 2024; the stability figure is the own-procedure half-sample refit reported in `results/CLAIMS_v1.17.md`.*

Post-freeze evidence extension — Gates 31–44

The first version of this report labelled the post-freeze experiments as proposed. They have now been implemented. Gates 31–44 are full runs; Gate 33 includes a 20-seed confirmation. This section records the results and the decisions they support. It is separate from Stages 1–12: the frozen configuration was not changed before these gates, and every gate has its own pre-registration under `results/`. A method is called rejected only when it was implemented, run on the same data and criteria, and failed those criteria.

Gate 32 — rule learners

Four alternatives to regression CART were implemented on the same discovery ψ, the same validation/assessment AIPW contrasts, the same support floor 200 and the same LCB commitment: cost-complexity pruned CART, PRIM, a classification tree on sign(ψ), and boosting distillation. The design is 20 fresh seeds per DGP (910100–910119), n = 8,000.

DGP	Learner	Purity	Capture	Jaccard	Commit	MAE	Cost (s)
single	CART (frozen)	0.889	0.531	0.487	80%	0.0556	0.06
single	pruned CART	0.741	0.642	0.452	85%	0.0562	2.94
single	PRIM	0.328	0.144	0.106	10%	0.0760	0.06
single	classification	0.671	0.576	0.442	70%	0.0603	0.05
single	boosting distill	0.883	0.543	0.477	75%	0.0621	105.1
two	CART (frozen)	0.811	0.379	0.296	65%	0.0729	0.02
two	pruned CART	0.637	0.640	0.362	85%	0.0762	1.62
two	PRIM	0.473	0.206	0.148	45%	0.0842	0.03
two	classification	0.701	0.403	0.280	75%	0.0737	0.02
two	boosting distill	0.761	0.287	0.239	65%	0.0699	25.1

Gate 32, 20 seeds per DGP. Cost is wall-clock seconds per replication. Full paired contrasts and confidence intervals: `results/gate32_verdict.md`.

Decision: keep regression CART. Pruned CART gains capture in the two-region DGP (+0.261, 95% CI [+0.089, +0.429]) but loses 0.173 purity; the pre-registered purity loss allowed is 0.02. In the single-region DGP its capture gain is not significant and it still loses 0.149 purity. The classification tree loses 0.219 / 0.110 purity and does not improve capture significantly. PRIM loses both capture and purity and needs 4.6–4.8 conditions. Boosting distillation is the only learner with a better two-region MAE (−0.0029), but it loses capture (−0.092) and purity

(−0.050) and costs 25–105 s per replication versus 0.02–0.06 s for CART, failing the 5× cost criterion by two orders of magnitude. This is a negative result for learner complexity, not a failure to search.

Gate 31 — outcome nuisance

Four outcome nuisances were implemented: per-arm Cox, random survival forest, survival gradient boosting, and equal-weight stacking. They share the same folds, censoring model, partitions and C200 learner. The design is 20 fresh seeds per DGP (900100–900119), 100-tree RSF and 60-estimator GBM.

DGP	Nuisance	MAE	Brier	Purity	Capture	Commit	Cost (s)
single	Cox	0.0517	0.0028	0.794	0.623	75%	0.002
single	RSF	0.0542	0.0073	0.802	0.615	70%	119.8
single	GBM	0.0528	0.0071	0.795	0.640	70%	147.5
single	stack	0.0475	0.0044	0.793	0.623	75%	260.4
two	Cox	0.0579	0.0033	0.833	0.424	80%	0.001
two	RSF	0.0560	0.0075	0.820	0.445	80%	77.2
two	GBM	0.0546	0.0075	0.797	0.401	75%	61.0
two	stack	0.0504	0.0048	0.803	0.433	80%	136.4

Gate 31, 20 seeds per DGP. Brier is the mean squared error of the predicted S(730) against the known truth. Full paired contrasts: `results/gate31_verdict.md`.

Decision: Cox is retained. No flexible model satisfies the joint criteria. Stacking is the only model with a significant MAE gain in both DGPs (−0.0043 single, −0.0075 two), but it loses 0.030 purity in the two-region DGP and costs 136–260 s per replication versus 0.002 s for Cox. GBM loses 0.036 purity in the two-region DGP. RSF is worse in the single-region DGP (+0.0025) and its two-region gain (−0.0019) is below the 0.002 threshold. Brier is worse for every flexible model, and coverage does not improve. This is a completed negative result for nuisance complexity, not a failure to search.

Gate 33 — calibration and intervals

Two questions were separated: how to calibrate the predicted leaf effect, and how to quantify uncertainty after the rule has been selected. Calibration is compared by assessment MAE; intervals are compared by coverage against the true selected-rule effect and by width. The main run uses 20 fresh seeds per DGP (920100–920119); a confirmation block on 921100–921119 is recorded separately.

DGP	Calibration	MAE	ΔMAE vs none	Purity	Commit
single	none	0.0531	—	0.915	90%
single	shrinkage	0.0523	−0.0009 [−0.0052, +0.0039]	0.915	90%
single	isotonic	0.0455	−0.0076 [−0.0121, −0.0034]	0.915	90%
two	none	0.0718	—	0.827	85%
two	shrinkage	0.0720	+0.0002 [−0.0032, +0.0033]	0.827	85%
two	isotonic	0.0684	−0.0034 [−0.0059, −0.0008]	0.827	85%

DGP	Interval	Coverage	Width	Width / IF
single	IF (frozen)	90%	0.203	1.00
single	selection bootstrap	95%	0.289	1.42
single	split-conformal	100%	0.492	2.42
two	IF (frozen)	90%	0.205	1.00
two	selection bootstrap	100%	0.274	1.34
two	split-conformal	100%	0.490	2.39

Gate 33 main run, 20 seeds per DGP. Full table: `results/gate33_verdict.md`.

The confirmation block uses 20 fresh seeds (921100–921119) and changes one decision.

DGP	Calibration	MAE	ΔMAE vs none [95% CI]	p
single	none	0.0522	—	—
single	shrinkage	0.0508	−0.0014 [−0.0067, +0.0040]	0.615
single	isotonic	0.0469	−0.0053 [−0.0104, −0.0001]	0.046
two	none	0.0727	—	—
two	shrinkage	0.0703	−0.0024 [−0.0066, +0.0019]	0.264
two	isotonic	0.0662	−0.0065 [−0.0099, −0.0032]	<0.001

DGP	Interval	Coverage	Width	Width / IF
single	IF (frozen)	95%	0.216	1.00
single	selection bootstrap	85%	0.270	1.25
single	split-conformal	100%	0.503	2.33
two	IF (frozen)	95%	0.228	1.00
two	selection bootstrap	95%	0.270	1.18
two	split-conformal	100%	0.531	2.33

Gate 33 confirmation, 20 fresh seeds. Purity, capture and commitment are identical across calibration methods.

Decision. Adopt isotonic calibration for prediction only (C200-I). It replicates in both DGPs, the paired CI excludes zero in both, and the policy is unchanged. Reject the selection bootstrap as the primary interval: its main-run advantage does not replicate (85% coverage in the single-region DGP, 95% in the two-region, both wider than IF). Reject split-conformal on width (2.33× IF). Keep the influence-function interval and report its measured coverage honestly. The SCRE head-to-head was run with C200-S; a fresh SCRE comparison with C200-I is not part of this gate.

Gate 34 — number of cross-fitting folds

Three, five and ten folds were compared on the same simulated data, holding the nuisances, target and partitions fixed. The design is 20 fresh seeds per DGP (930100–930119).

DGP	Folds	Bias	Coverage	MAE	ESS	Cost (s)	ΔMAE vs 5
single	3	+0.0023	95%	0.0522	2,798.7	6.7	+0.0011
single	5	+0.0022	95%	0.0511	2,798.9	12.4	—
single	10	+0.0023	95%	0.0506	2,799.0	28.2	−0.0005
two	3	+0.0018	95%	0.0586	2,834.8	8.8	+0.0010
two	5	+0.0017	95%	0.0576	2,834.9	15.6	—
two	10	+0.0018	95%	0.0571	2,835.0	32.8	−0.0005

Decision: keep 5 folds. Ten folds improves MAE by 0.0005 — real but one quarter of the pre-registered 0.002 threshold — at 2.1–2.3× the cost. Three folds is worse by 0.0010–0.0011. Coverage is 95% in every cell. The fold count is therefore a tested design compromise, not an unexamined default.

Gate 35 — fairness, SHAP and internal transport

This gate is descriptive: ACTG175 has no individual treatment-effect truth. The main C200 rule reproduces the canonical result exactly: `cd40 ≤ 407.5 & wtkg > 71.0 & homo ≤ 0.5`, discovery support 61, validation +0.116 (LCB −0.250) → INSUFFICIENT, assessment +0.401 [+0.102, +0.700] on 38 patients.

Group	n	Rule n	Prevalence	Effect	95% CI	Powered
white	183	24	13.1%	+0.444	[+0.081, +0.807]	no

Group	n	Rule n	Prevalence	Effect	95% CI	Powered
non-white	81	14	17.3%	+0.328	[−0.191, +0.847]	no
male	213	25	11.7%	+0.514	[+0.147, +0.881]	no
female	51	13	25.5%	+0.184	[−0.318, +0.686]	no

Gate 35, assessment partition. Every group has fewer than 30 rule patients; the intervals overlap. No fairness claim is made.

Calibration at 730 is good overall (ECE 0.023 white, 0.040 non-white, 0.001 male, 0.058 female); the female stratum is the worst and over-predicts treated-arm survival by 0.094. SHAP on a surrogate for ψ ranks `wtkg` (0.048), `cd40` (0.046), `cd80` (0.042), `str2` (0.017) and `homo` (0.011); the rule uses `wtkg`, `cd40` and `homo`, so the surrogate is consistent with the rule but is not the explanation. The internal white→non-white transport rule is `cd40 ≤ 415.5 & homo ≤ 0.5`, support 74 (<200, fallback); its effect is +0.236 [−0.053, +0.525] on white assessment and +0.223 [+0.071, +0.375] on non-white. This is a feasibility signal, not external validation.

Decision: no model change; three limitations added. Fairness cannot be established from 38 rule patients in assessment; the female stratum is the worst calibrated; and internal transport loses the support floor. These are reported, not tuned away.

Gate 36 — AIPW vs fixed-horizon IPCW-TMLE

The last proposed estimator was implemented as a fixed-horizon IPCW-TMLE: the outcome is the binary event-free indicator at 730 days, the clever covariate is H_a = 1{A=a}/e_a × Y/G, and the initial Cox nuisances are updated by a logistic fluctuation. It uses the same folds, partitions and C200 learner as AIPW. This is not the longitudinal TMLE over the whole survival curve.

DGP	Estimator	Bias	Coverage	MAE	Brier	Purity	Capture	Commit
single	AIPW	−0.0014	95%	0.0516	0.0026	0.869	0.582	70%
single	TMLE	−0.0052	100%	0.1408	0.0177	0.932	0.542	85%
two	AIPW	−0.0019	95%	0.0578	0.0031	0.836	0.390	80%
two	TMLE	−0.0111	100%	0.1374	0.0172	0.904	0.398	90%

Gate 36, 20 seeds per DGP (940100–940119). Cost: 4.5 s for TMLE vs effectively zero for AIPW. Full table: `results/gate36_verdict.md`.

Decision: AIPW is retained; TMLE is rejected. TMLE is purer (+0.064 / +0.068) and has better assessment MAE (−0.010 / −0.005), but patient-level MAE is 0.089 / 0.080 worse, Brier is 0.015 / 0.014 worse, absolute bias is larger, capture falls 0.039 in the single-region DGP, and cost is 4.5 s vs effectively zero. It fails the pre-registered MAE/bias/capture/cost criteria. The targeting step moves the marginal estimate away from the truth and distorts the patient-level contrast. This is a negative result for this fixed-horizon TMLE, not for longitudinal TMLE.

Gate 37 — horizon selection

The horizon had been fixed at 730 days on clinical grounds and examined only by sensitivity analysis. Gate 37 turns that into a selection check: 365, 548, 730, 912 and 1,095 days are compared on events, patients still at risk, support, validation precision and the selected rule. Assessment is descriptive only and is not used to choose the horizon.

Horizon	Events	At risk	Support	Val. effect	Val. SE	Val. LCB	Assessment
365	46	87.5%	77	+0.171	0.091	+0.021	+0.017
548	69	80.6%	109	+0.219	0.108	+0.042	+0.083
730	99	70.8%	61	+0.116	0.222	−0.250	+0.401
912	135	57.9%	61	+0.348	0.239	−0.044	+0.319
1,095	143	21.8%	237	−0.162	0.215	−0.516	+0.575

Gate 37, ACTG175, discovery/validation only. Rules and full output: `results/gate37_horizon.csv`.

Decision: keep 730 days. No horizon satisfies all four pre-registered criteria (≥150 events, ≥30% at risk, validation SE within 0.02 of 730, support ≥200 or a documented fallback ≥60). 365 and 548 have positive LCBs but only 46 and 69 events; 1,095 has 51% censored and only 22% at risk; 912 has 135 events but its LCB is −0.044. The 730-day choice was initially a documented compromise; Gate 44 later shows that it is the most stable horizon across 20 splits (mean validation LCB −0.142, SD 0.114, versus 845 and 965).

Gate 38 — feature selection: 12 vs 16, lasso and RFE

The 12 published covariates were retained for comparability with SCRE. Gate 38 tests whether the 16-candidate set, lasso or RFE selects a feature set that improves the rule. Feature selection is fitted on discovery only; the target and nuisances are refitted on the selected columns.

Method	Features	Support	Fallback	Val. effect	Val. SE	Val. LCB	Conditions
12	12	61	yes	+0.116	0.222	−0.250	3
16	16	61	yes	+0.110	0.223	−0.256	3
lasso	12	61	yes	+0.116	0.222	−0.250	3
RFE	11	79	yes	+0.148	0.180	−0.148	2

Gate 38, ACTG175. Lasso selects all 12 features. RFE drops `age` and selects `cd80 > 805.5 & race > 0.5`, support 79. Full output: `results/gate38_features.csv`.

Decision: keep the 12 covariates. Lasso selects all 12, so it changes nothing. RFE drops `age`, replaces the rule entirely and falls to support 79 (<200, fallback). Its validation SE is smaller, but the rule is a different subgroup below the support floor; it does not meet the pre-registered adoption criteria. The 12-feature set is retained for comparability and interpretability, with the RFE result reported as a sensitivity.

Gate 39 — learned stacking meta-learner

Gate 31 tested equal-weight stacking. Gate 39 fits positive weights on Cox, RSF and GBM out-of-fold predictions against the IPCW outcome. 10 seeds per DGP.

DGP	Estimator	MAE	Purity	Capture	Commit	Assessment MAE
single	Cox	0.0505	0.844	0.560	80%	0.0524
single	equal weight	0.0467	0.885	0.524	80%	0.0561
single	learned	0.0513	0.869	0.551	80%	0.0539
two	Cox	0.0573	0.817	0.413	80%	0.0692
two	equal weight	0.0509	0.817	0.413	90%	0.0689
two	learned	0.0573	0.817	0.413	80%	0.0695

Gate 39, 10 seeds per DGP. Mean weights: Cox 0.77 / 0.76, RSF 0.14 / 0.23, GBM 0.09 / 0.02.

Decision: keep Cox. The learned stack is indistinguishable from Cox (+0.0007 / +0.0001 MAE, ns) because the weights put ~0.77 on Cox. Equal-weight stacking lowers MAE (−0.0038 / −0.0064) but loses 0.036 capture in the single-region DGP; its purity gain is a different operating point, not a dominant one. It is a sensitivity, not adopted.

Gate 40 — SCRE full alpha path

The head-to-head uses one SCRE alpha path chosen by validation concordance. Gate 40 fits one SCRE model per replication and evaluates 12 alpha points on the same assessment patients.

DGP	Best MAE	Chosen MAE	Gap	Best purity	Chosen purity	Best capture	Chosen capture
single	0.0558	0.0568	+0.0010	0.421	0.410	0.968	0.983
two	0.0687	0.0704	+0.0017	0.554	0.600	0.842	0.806

Gate 40, 10 seeds per DGP. Path ranges: single MAE 0.056–0.079, purity 0.345–0.442, capture 0.911–0.976; two MAE 0.069–0.090, purity 0.477–0.605, capture 0.769–0.904.

Decision: keep the validation-concordance alpha. It is close to the best-MAE alpha (gap 0.0010–0.0017), and the path does not overturn the head-to-head direction. The exact purity/capture numbers are path-sensitive; the conclusion that SCRE is more accurate and C200 purer is not. Selecting the best-MAE alpha post hoc would use the assessment partition and is rejected.

Gate 41 — selection-aware intervals

The IF interval conditions on the selected rule. Gate 41 tests two practical selection-aware alternatives: Bonferroni over candidate leaves, and sample-splitting the discovery set into selection and estimation halves.

DGP	Method	Coverage	Width	Width / IF
single	IF	95%	0.298	1.00
single	Bonferroni	100%	0.412	1.38
single	sample-splitting	100%	0.298	1.00
two	IF	100%	0.297	1.00
two	Bonferroni	100%	0.412	1.39
two	sample-splitting	100%	0.294	0.99

Decision: keep the IF interval as primary. It already reaches 95–100% coverage and is narrower than Bonferroni. Report Bonferroni and sample-splitting as validated selection-aware sensitivities. The IF interval still conditions on the selected rule; the limitation is quantified, not removed. Sample-splitting reports a rule selected on half of discovery, so it is a different estimand from the full-discovery C200 rule.

Gates 42–44 — horizon optimisation and stability

Gate 37 retained 730 as a compromise. Gates 42–44 ask whether another horizon is actually better. Gate 42 searches 365–1,095 in 30-day steps and maximises validation LCB; Gate 43 selects by discovery-only inner cross-validation; Gate 44 repeats the comparison over 20 random splits to break the tie. No assessment value is used to choose the horizon.

Horizon	Mean val. LCB	SD	Mean assessment	Commit	Single-split val. LCB	Discovery inner LCB
730	−0.142	0.114	+0.206	5%	−0.250	−0.118 (725)
845	−0.167	0.185	+0.251	15%	−0.200	+0.014
965	−0.206	0.191	+0.220	10%	−0.000	−0.265

Gate 44, 20 splits, seeds 2024–2043. Single-split values from Gate 42; discovery inner LCB from Gate 43. Full tables: `results/gate44_summary.csv`, `gate42_horizon_grid.csv`, `gate43_horizon_inner.csv`.

Finding: the criteria disagree on one split. Validation prefers 965 (LCB −0.000), discovery-only inner CV prefers 845 (+0.014), and 730 is the clinical default. Across 20 splits, 730 has the best mean validation LCB (−0.142) and the smallest SD (0.114); 845 has a higher mean assessment effect (+0.251) but worse LCB (−0.167) and 1.6× the variability; 965 is worst on both mean LCB and SD. The single-split advantages do not reproduce.

Decision: keep 730 days. This is not a defence of the old value; it is the 20-split stability result. 845 remains a reported sensitivity with a higher mean assessment effect, but it is less stable and commits less often. No horizon meets the commitment threshold on average; the model abstains under all three. Gate 37's earlier 'compromise' wording is replaced by this stability result.

Threats to validity

A threat is listed only if it could change a reported conclusion. Each threat is assigned to a type, given its measured evidence, and marked as controlled, disclosed or open. 'Disclosed' means the limitation is reported but not repaired; 'open' means the experiment needed to settle it has not been run.

Type	Threat	Evidence	Consequence	Response	Status
Internal	Leakage in hyperparameter tuning	Naive CV 0.824 vs nested 0.8638 for the same criterion	An apparent gain that is an artefact of tuning on a precomputed ψ	Refit nuisances inside every fold; withdraw the claim	Controlled
Internal	Selection on the validation partition	Intervals condition on the selected rule; IF coverage 90–92.5%	Nominal 95% intervals are slightly optimistic	Commit only on the LCB; report measured coverage	Disclosed
Internal	Winner's curse at discovery	Largest leaf 173 < 200 on ACTG175	The reported leaf effect can be larger than the true effect	Support floor plus independent validation	Partially controlled
Construct	Fixed 730-day composite endpoint	$\hat{\tau}$ and the selected rule change at 365 and 1,095 days	Conclusions are horizon-specific	Horizon fixed before analysis; sensitivity reported	Disclosed
External	ACTG175 population and era	Men, 1990s regimens, 1,054 patients	No transportability to current or female populations	External validation proposed, not run	Open
Statistical	Small discovery sample	20 distinct rules from 20 splits; Jaccard ≈ 0.34	The exact rule is not reproducible	Report instability; do not tune until it looks stable	Open
Censoring	Conditional independent censoring	Cannot be tested; δ-tilt keeps the sign (+0.401 to +0.456)	Informative dropout would bias τ	State the assumption; sensitivity analysis	Open
Comparator	One SCRE α path	Validation concordance returns roughly one rule	The head-to-head does not cover the full SCRE path	Describe the comparison, not SCRE in general	Disclosed
Generalisation	Two simulated DGPs, one covariate matrix	Single- and two-region effects on ACTG175 X	No claim beyond this family	Label the scope explicitly	Open

Reproducibility manifest

Item	Record
Raw data	`ACTG175.csv` (project root); SHA-256 `debab27bbc6f741724b79e49a8e13e1feb8db57850b17010b0635202bfc7141a`
Analysis population	arms 0/1, n = 1,054, 12 baseline covariates
Notebook	`notebooks/CISCaRL_31_steps.ipynb`, 136 cells (52 code), executed end to end with zero error cells
Library	`ciscarl2/` (estimators, gates, policy evaluation, diagnostics, nested tuning); 70 modules compile with zero import failures
Python	3.11.0
Packages	numpy 2.4.6; pandas 2.2.3; scikit-learn 1.9.1; scipy 1.17.1; lifelines 0.30.3; matplotlib 3.11.2; statsmodels 0.14.6
Split seeds	2024 (discovery vs holdout); 2025 (validation vs assessment)
Head-to-head seeds	300100–300139 (× two DGPs = 80 paired replications)
Tuning seeds	310100–310139 (Gate 30 knob sweep)
Confirmation seeds	320100–320139 (disjoint from tuning)
Sweep seeds	1,000 descriptive replications per design; no confirmatory block
Policy seeds	700100 (40 paired replications)
Split-ratio seeds	4000–4019 per ratio (20 real splits)

Item	Record
Canonical results	`results/CANONICAL_RESULTS.md` — the single source of truth for every number in this report
Claims	`results/CLAIMS_v1.17.md` — defensible claims and withdrawn claims
Diagnostics	`results/diagnostics.md` plus the four `diagnostics_*.csv` files
Audit	`results/FINAL_AUDIT.md` and `results/AUDIT_report.md`; the audit passes before any experiment is quoted
Paper	`paper/CISCaRL_IEEE_conference_detailed.docx` / `.pdf`; 8 pages, 15 figures, 10 tables; builder `paper/build_ieee_detailed.py`
Version control	Working tree only; no commit hash is recorded in this report, so the file hashes and seed blocks are the reproducibility anchor
Re-run	`python run_all.py` (full evidence), `--quick` (sandbox smoke test), `--dry-run` (command list); environment pinned in `requirements.txt`

Experiment register

ID	Stage	Experiment	Design	Status
E1	2	27-column audit	descriptive counts	Completed
E2	3	16-covariate sensitivity	refit, same split	Completed
E3	4	Split-ratio study	4 ratios × 20 splits + 20 sim reps	Completed
E4	4	A×E vs A stratification	event-rate spread	Completed
E5	5	PH diagnostics	Schoenfeld, per arm	Completed
E6	5	Efron vs Breslow	coefficient comparison	Completed
E7	5	Survival calibration	predicted vs KM, 3 bins/arm	Completed
E8	5	Weight-cap sweep	1.2 to 20	Completed
E9	3/6	Sample-size boundary	n = 2,000–6,000, 40 seeds	Completed
E10	7	Nested-refit tuning	5 outer folds, 18 configs	Completed
E11	7	Two-leaf list	40 seeds	Rejected
E12	7	Flexible learner + distillation	40 seeds	Rejected
E13	8	Support floor sweep	none/100/200/400/prop	Completed
E14	8	Shrinkage calibration	80 paired reps + confirm	Completed
E15	8	Interval coverage	40 reps, rule contrast	Completed
E16	10	Fresh-seed confirmation	40 seeds	Completed
E17	11	C200 vs SCRE	80 paired reps	Completed
E18	11	1,000-rep sweeps	two designs	Completed
E19	11	Policy value and net benefit	40 paired reps	Completed
E20	12	Horizon sensitivity	365/730/1,095	Completed
E21	12	Missingness stress	5% in cd40, median impute	Completed
E22	12	Bootstrap stability	own-procedure refits	Completed
E23	5	Flexible nuisance: Cox vs RSF / GBM / stack (Gate 31)	20 seeds × 2 DGPs	Completed
E24	8	Selection bootstrap / conformal interval (Gate 33)	20 seeds × 2 DGPs	Completed
E25	12	Fairness, SHAP, internal transport (Gate 35)	ACTG175 assessment	Completed
E26	12	External validation (PBC)	out of scope: ACTG175 only	Not run
E27	7	Rule learners: pruned, PRIM, classification, boosting (Gate 32)	20 seeds × 2 DGPs	Completed
E28	8	Isotonic vs shrinkage calibration (Gate 33)	main + 20-seed confirmation	Completed
E29	5	Censoring model: marginal vs conditional	Gate 18, 40 seeds, κ = 0 / 3	Completed
E30	6	Estimator alternatives: pooled Cox, T-learner, IPW, plug-in	Gate 1c / 9 / 17	Completed
E31	4	Fold count 3 / 5 / 10 (Gate 34)	20 seeds × 2 DGPs	Completed
E33	6	AIPW vs fixed-horizon IPCW-TMLE (Gate 36)	20 seeds × 2 DGPs	Completed
E34	5	Horizon selection 365–1,095 (Gate 37)	ACTG175 discovery/validation	Completed
E35	3	Feature selection 12 / 16 / lasso / RFE (Gate 38)	ACTG175 discovery	Completed
E36	11	SCRE alpha path (Gate 40)	10 seeds × 2 DGPs, 12 alphas	Completed

ID	Stage	Experiment	Design	Status
E37	8	Selection-aware intervals (Gate 41)	20 seeds × 2 DGPs	Completed
E38	5	Learned stacking meta-learner (Gate 39)	10 seeds × 2 DGPs	Completed
E39	5	Horizon optimisation and stability (Gates 42–44)	dense grid + 20 splits	Completed

Decision register

ID	Decision	Evidence	Type
D1	Arms 0/1, composite endpoint, 730 days	Trial protocol; 284 events; 71% at risk	Design
D2	Treatment-policy estimand	390 off-treatment, 122 events, follow-up to 1,176 d	Design
D3	12 baseline covariates	zprior constant; z30 94.8%; oprior 2.4%; comparability	Design
D4	Keep and flag implausible values	3 cd40=0; 199 outside range; 13 minors	Design
D5	50/25/25, stratified A×E	Event spread 0.5 vs 2.8; 4-ratio study	Experimental
D6	Scaler fitted on discovery	Leakage control; centring required by Cox baseline	Design
D7	Cox nuisances, 5-fold, cap min(Q99,20)	PH global p=0.24/0.37; ESS 372; cap benign	Experimental
D8	AIPW, e = 0.5	Audit bias +0.0031	Design
D9	Depth 3 / leaf 60 / squared error	Nested CV 0.8638 vs best 0.8633	Experimental
D10	Support floor 200 (proportional below 4,000)	Capture 0.148→0.364; purity 0.77→0.82	Experimental
D11	Commit on LCB > 0	0% null; efficiency 0.82–0.88	Experimental
D12	Shrinkage for prediction only (superseded by D23)	ΔMAE −0.0051/−0.0026; policy unchanged	Superseded
D13	Absolute-error criterion	Withdrawn: 0.8711 nested vs 0.8638	Rejected
D14	Two-leaf list	Purity 0.862→0.696	Rejected
D15	Flexible learner + distillation	Capture 0.548→0.454	Rejected
D16	One benchmark, two-sided reporting	Tie after calibration; precision vs coverage	Experimental
D17	Policy value separated from net benefit	V(d) vs U(d)=V−λC	Design
D18	Intervals are not conformal	No conformal procedure implemented	Design
D19	Report 90–95% IF coverage, not 95% as a guarantee	40-rep policy audit + Gate 33 confirmation	Experimental
D20	Regenerate paper on A×E split	Canonical result record	Design
D21	Keep regression CART	Gate 32: no learner meets purity/cost guardrails	Experimental
D22	Keep 5 cross-fitting folds	Gate 34: 10 folds gains 0.0005 MAE, below 0.002	Experimental
D23	Isotonic calibration for prediction	Gate 33 confirmation: −0.0053 / −0.0065 MAE	Experimental
D24	Reject selection bootstrap; keep IF	Gate 33 confirmation: 85% / 95% coverage	Experimental
D25	No fairness or transport claim	Gate 35: every group < 30 rule patients	Design
D26	Keep Cox; reject flexible nuisances	Gate 31: stack loses 0.030 purity and costs 136–260 s; GBM loses 0.036 purity	Experimental
D27	Keep AIPW; reject fixed-horizon IPCW-TMLE	Gate 36: TMLE MAE 0.079–0.089 worse, Brier worse, cost 4.5 s	Experimental
D28	Keep 730-day horizon	Gate 44: best mean validation LCB (−0.142) and lowest SD across 20 splits	Experimental
D29	Keep 12 covariates	Gate 38: lasso = 12; RFE changes rule, support 79	Experimental
D30	Keep SCRE validation-concordance alpha	Gate 40: chosen within 0.0017 MAE of best; path does not overturn direction	Experimental
D31	Keep IF interval; report Bonferroni/sample-split sensitivities	Gate 41: IF 95–100%, Bonferroni 1.38× width	Experimental
D32	Keep Cox; reject learned stacking	Gate 39: learned stack = Cox (ns); equal weight loses capture	Experimental

Implementation status

Component	Status	Evidence
12-stage pipeline	Documented	this report
Cohort and 12 covariates	Completed	1,054 patients; 27-row audit
Split and leakage	Completed	notebook Steps 4 and 7
PH / tie / calibration / cap diagnostics	Completed	results/diagnostics.md
Five-fold cross-fitting	Completed	fold table; audit
Nested nuisance refitting in tuning	Completed	nested_tuning.csv
Regression CART	Completed	6 leaves; rule
Support selection	Completed	floor sweep
Validation and commitment	Completed	LCB −0.250; INSUFFICIENT
Calibration (isotonic; C200-S used in SCRE)	Completed	Gate 33 confirmation
Interval coverage audit	Completed	IF 90–95%; selection bootstrap rejected (85% / 95%)
Policy value and net benefit	Completed	policy_eval_summary.csv
Assessment at Step 23	Completed	notebook re-ordered
Paper–notebook reconciliation	Completed	CANONICAL_RESULTS.md
Estimator alternatives (pooled Cox, T-learner, IPW, plug-in)	Completed	Gate 1c / 9 / 17
Censoring-model alternatives (marginal vs conditional)	Completed	Gate 18
Flexible nuisance: Cox vs RSF / GBM / stack	Completed	Gate 31; Cox retained
Rule learners: pruned, PRIM, classification, boosting	Completed	Gate 32; CART retained
Isotonic calibration	Completed, adopted	Gate 33 confirmation
Selection bootstrap / conformal inference	Completed, rejected	Gate 33 confirmation; IF retained
Fairness, SHAP, internal transport	Completed	Gate 35; underpowered
Fold count 3 / 5 / 10	Completed	Gate 34; 5 retained
Horizon 365–1,095	Completed	Gate 44; 730 best across 20 splits
Feature selection 12 / 16 / lasso / RFE	Completed	Gate 38; 12 retained
SCRE alpha path	Completed	Gate 40; direction stable
Selection-aware intervals	Completed	Gate 41; IF retained
Learned stacking meta-learner	Completed, rejected	Gate 39; learned stack = Cox; equal weight loses capture
AIPW vs fixed-horizon IPCW-TMLE	Completed, rejected	Gate 36; AIPW retained
External validation (PBC)	Out of scope	ACTG175 only

Notebook mapping

Stage	Notebook step	What runs there
1	1–3	environment, audit, cohort, endpoint
2	2	27-row role audit, EDA
3	3–6	cohort, feature selection, preprocessing
4	4	split and split-ratio study
5	8–9	nuisances, diagnostics
6	9–10	AIPW ψ, nested tuning

Stage	Notebook step	What runs there
7	10, 18	CART, nested tuning
8	11, 15	selection, validation, commitment, calibration
9	19	configuration and decision log
10	20–23	confirmation, assessment
11	21–22	evaluation, SCRE, policy value
12	24, 31	ablations, packaging
Gate 31	16	nuisance comparison (full 20 seeds)
Gate 36	16	AIPW vs fixed-horizon IPCW-TMLE
Gate 37	24	horizon selection
Gates 42–44	24	horizon optimisation and 20-split stability
Gate 38	24	feature selection
Gate 40	22	SCRE alpha path
Gate 41	26	selection-aware intervals
Gate 32	17	rule learners (full 20 seeds)
Gate 33	26	calibration and intervals (full main run)
Gate 34	27	fold count (full 20 seeds)
Gate 35	28	fairness, SHAP, internal transport
Summary	29–30	evidence summary and decision register

Final checklist before submission

- Every number in the paper matches results/CANONICAL_RESULTS.md.
- The notebook executes end to end with zero error cells.
- The paper compiles with no undefined references and no overfull boxes.
- The audit passes before any experiment is quoted.
- Proposed experiments are labelled as proposed, not reported as results.
- The withdrawn absolute-error claim does not appear as evidence.
- The 90–95% IF coverage range is stated wherever an interval is reported.
- The ACTG175 section reports abstention, not a subgroup recommendation.
- The repository, data and seeds are sufficient to reproduce the tables.